Lecture #8, Oct 7, 2025

Explicit scheme for transport

Last lecture: multi stage scheme
multi step scheme
non-linear system

Lorenz system for convection

$$\frac{dx}{dt} = \sigma(y-x)$$

$$\frac{dy}{dt} = x(\rho-z)-y$$

$$\frac{dz}{dt} = xy-\beta z$$

Here, x is intensity of convection (rate of fluid flow)
y is temperature difference
z is distortion of vertical temperature

It is a chaotic system and multi stage method is a good choice.

Lorenz system is a model of atmospheric convection.

The standard PDE form of convection or transport equation is

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

1D wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

$$\frac{\partial u}{\partial t} - c\frac{\partial u}{\partial x} = 0$$

$$a^2 - b^2 = 0$$

$$(a+b)(a-b) = 0$$

This is a second order hyperbolic PDE

A first order hyperbolic PDE is

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0, \quad c > 0$$

This linear hyperbolic PDE describes a wave propagating in the x direction with a speed c.

Its solution is

$$u(x,t) = F(x-ct)$$

The initial solution $u(x,0) = F(x)$ remains constant over time along the line $x - ct = 0$ (characteristic line)

Discretization

time derivative

$$\frac{\partial u}{\partial t} = \frac{u_j^{n+1} - u_j^n}{\Delta t} + O(\Delta t)$$

where $u(x_j, t_n) \equiv u_j^n$

$$\Delta t = t^{n+1} - t^n$$

space derivative

$$\frac{\partial u}{\partial x} = \frac{u_{j+1}^n - u_j^n}{\Delta x} \quad \text{forward}$$

$$\frac{\partial u}{\partial x} = \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} \quad \text{centered}$$

$$\frac{\partial u}{\partial x} = \frac{u_j^n - u_{j-1}^n}{\Delta x} \quad \text{backward}$$

Euler explicit scheme with backward in space

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c\,\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0, \quad c>0 \quad (EEB)$$

lead error $O(\Delta t, \Delta x)$ (upwind method / upstream method)

Euler explicit scheme with centered in space

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c\,\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0 \qquad \text{(EEC)}$$

lead error $O(\Delta t, \Delta x^2)$

Leap-Frog scheme with centered in space

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + c\,\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0, \qquad O(\Delta t^2, \Delta x^2)$$

Ex: Write down Euler explicit scheme for 1D advection with $a > 0$ and backward in space:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\,\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0$$

Show that the stability condition is

$$\frac{a\Delta t}{\Delta x} \leqslant 1 \qquad \left(\begin{array}{c} \text{stability E.E.} \\ |(1+\lambda\Delta t)| \leq 1 \end{array} \right)$$

H.W.

Remark This is called CFL condition

Take $\text{CFL} = \dfrac{a\Delta t}{\Delta x}$, then $\boxed{\Delta t = \text{CFL}\,\dfrac{\Delta x}{a}}$

Consistency:

show that Forwar Euler scheme with upwind discritization is consistent.

Sol^n $\qquad \dfrac{\partial u}{\partial t} + a\dfrac{\partial u}{\partial x} = 0, \qquad a>0$

$$\frac{\partial u}{\partial t} = \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \cdots \qquad (1)$$

$$\frac{\partial u}{\partial x} = \frac{u_j^n - u_{j-1}^n}{\Delta x} + \frac{\Delta x}{2}\frac{\partial^2 u}{\partial x^2} - \cdots \qquad (2)$$

(1) + a (2) = P

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} - \left[\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\,\frac{u_j^n - u_{j-1}^n}{\Delta x}\right] = -\frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{a\Delta x}{2}\frac{\partial^2 u}{\partial x^2} + \cdots$$

Eq^n $\quad - \quad$ scheme $\quad = \quad$ Error

$= 0$

As $\Delta t \to 0$, $\Delta x \to 0$, Error $= -\dfrac{\Delta t}{2}\dfrac{\partial^2 u}{\partial t^2} + \dfrac{a\Delta x}{2}\dfrac{\partial^2 u}{\partial x^2} \to 0$

Lecture #9, Oct 9, 2025

Von Neumann stability and

modified equation

Lax theorem: $\Rightarrow$ stability is necessary and sufficient for Convergence (Lect #1)

Von Neumann $\Rightarrow$ If error decay and damp out then the scheme is stable.

Ex: ⓐ Show that the scheme

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\,\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0$$

is stable if $\frac{a\Delta t}{\Delta x} \le 1$

u_j^n is approximation on (x_j, t_n)

$u(x_j, t_n)$ is exact solⁿ

(b) let $\varepsilon_j^n = u(x_j, t_n) - u_j^n$

For any numerical scheme of a linear PDE, error satisfies the same scheme.

In (a), we have

$$\frac{\varepsilon_j^{n+1} - \varepsilon_j^n}{\Delta t} + a\,\frac{\varepsilon_j^n - \varepsilon_{j-1}^n}{\Delta x} = 0$$

$\underline{\text{Sol}^n}$ Take Fourier mode $e^{i(kx+\omega t)}$

and replace as $u_j^n = e^{i(kx_j+\omega t_n)}$, k real, ω complex

Then $u_{j-1}^n = e^{i(kx_j - k\Delta x + \omega t_n)}$

$$= e^{-ik\Delta x} \underbrace{e^{i(kx_j+\omega t_n)}}_{u_j^n}$$

$$= e^{-ik\Delta x} u_j^n$$

$$\boxed{\begin{aligned} u_j^{n+1} &= e^{i(kx_j+\omega t_n+\omega\Delta t)} \\ &= e^{i\omega\Delta t} e^{i(kx_j+\omega t_n)} \end{aligned}}$$

not be using

Plug back:

$$\frac{u_j^{n+1}-u_j^n}{\Delta t} + a\frac{u_j^n - e^{-ik\Delta x}u_j^n}{\Delta x} = 0$$

Multiply by Δt and devide by u_j^n

$$\frac{u_j^{n+1}}{u_j^n} - 1 + \frac{a\Delta t}{\Delta x}\left(1-e^{-ik\Delta x}\right) = 0$$

Let $G = \frac{u_j^{n+1}}{u_j^n}$, $\beta = \frac{a\Delta t}{\Delta x}$ (CFL number)

$$\theta = k\Delta x$$

$$G = 1 - \beta(1-\cos\theta + i\sin\theta)$$

$$G = 1-\beta + \beta(\cos\theta - i\sin\theta)$$

$$|G|^2 = (1-\beta+\beta\cos\theta)^2 + \beta^2\sin^2\theta$$

$$= (1-\beta)^2 + 2(1-\beta)\beta\cos\theta + \beta^2$$

$$|G|^2 = 1 - \underbrace{2\beta(1-\beta)(1-\cos\theta)}_{<1 \text{ if } \beta<1}$$

Hence $|G| \le 1$ if $\beta \le 1$ or $\boxed{\frac{a\Delta t}{\Delta x} \le 1}$

or the scheme is conditionally stable

(b) H.W

Remark. i) $G = \dfrac{u_j^{n+1}}{u_j^n}$ is calle amplification factor.

$\Rightarrow$ the growth of $\dfrac{\varepsilon_j^{n+1}}{\varepsilon_j^n}$

ii) $u_j^n = e^{i(kx_j + \omega t_n)}$, k real, ω comple.

$\omega = Re\,\omega + i\,Im\,\omega$

$$u_j^n = e^{-Im\,\omega\, t_n}\, e^{i(kx_j + Re\,\omega\, t_n)}$$

$$\frac{u_j^{n+1}}{u_j^n} = e^{-Im\,\omega\,\Delta t}$$

H.W. show

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\,\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

is unconditionally unstable.

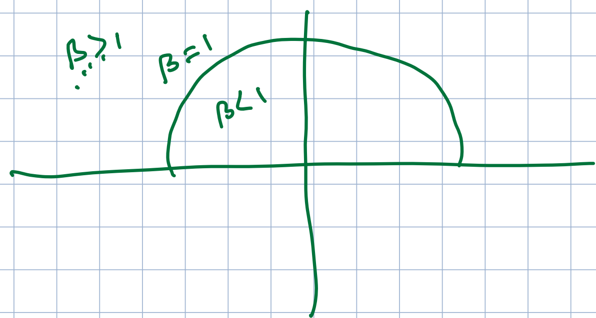
Also, $\dfrac{u_j^{n+1} - u_j^n}{\Delta t} + a\,\dfrac{u_{j+1}^n - u_j^n}{\Delta x} = 0, \quad a > 0$

is unconditionally unstable.

H.W

$G = 1 - \beta + \beta(\cos\theta - i\sin\theta), \quad \beta = \dfrac{a\Delta t}{\Delta x}, \quad \theta = k\Delta x$

Plot G is some suitable complex plane.



H.W.

From Lec # 6 (PS 9-10)

$G(\lambda\Delta t) = 1 + \lambda\Delta t$ for Euler explicit.

Show $\lambda = -\dfrac{a}{\Delta x}(1 - e^{-ik\Delta x})$

for Euler exp. Scheme of 1D advection (wave) eq^n

using upwind scheme.

Hint $\dfrac{du_j}{dt} + a\,\dfrac{u_j - u_{j-1}}{\Delta x} = 0$

Set $u_j = e^{i(kx_j + \omega t)}$

H.W.

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0, \quad a > 0 \qquad \text{Euler exp. + upwind.}$$

Show that RK4 is stable for

CFL, $\boxed{\frac{a\Delta t}{\Delta x} \le 1.39}$

Hint Let $z = \lambda \Delta t$

$$G(\lambda\Delta t) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$$

$$= e^z - \underbrace{\frac{z^5}{5!} - \cdots\cdots}$$

$$\approx e^z + O(\Delta t^5)$$

If $\lambda\Delta t = iy$ for a scheme

$$G(\lambda\Delta t) = e^{iy} + O(y^5)$$

Remark. RK4 does not improve the convergence if Euler exp + upwind is used for wave eqn.

Modified Equation analysis

Consider $\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0, \quad a > 0$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\,\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0, \qquad \text{Euler exp + upwind}$$

Taylor series,

Follow Lect #7

$$\frac{\partial u}{\partial t} = \frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \cdots \qquad (1)$$

$$\frac{\partial u}{\partial x} = \frac{u_j^n - u_{j-1}^n}{\Delta x} + \frac{\Delta x}{2}\frac{\partial^2 u}{\partial x^2} - \cdots \qquad (2)$$

(1) + a (2) = 0

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} - \underbrace{\left[\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\frac{u_j^n - u_{j-1}^n}{\Delta x}\right]}_{=0 \text{ by scheme}} = \underbrace{-\frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta x}{2}\frac{\partial^2 u}{\partial x^2}}_{\text{lead error}} + \cdots$$

$$\Rightarrow \quad \frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = -\frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta x\, a}{2}\frac{\partial^2 u}{\partial x^2}$$

This is called modified eq^n which is solved by the scheme.

Remove time derivative from R.H.S.

then

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = -\frac{\Delta t}{2} a^2 \frac{\partial^2 u}{\partial x^2} + \frac{\Delta x\, a}{2}\frac{\partial^2 u}{\partial x^2}$$

$$= \underbrace{\frac{a\Delta x}{2}\left(1 - \frac{a\Delta t}{\Delta x}\right)\frac{\partial^2 u}{\partial x^2}}_{\text{Second order term in 1st order error.}}$$

will be decaying (see HW.1)

if $\frac{a\Delta t}{\Delta x} \leq 1$

End of week 5 lectures

Below are some additional notes which may be helpful.

4. [Week 5]

(a) Show that the modified equation of the following finite difference scheme

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0$$

is

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = \frac{a\Delta x}{2}(1-\nu)\frac{\partial^2 u}{\partial x^2}$$

Note that the wave equation is

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0.$$

Differentiate the wave equation with respect to t to get

$$u_{tt} + au_{xt} = 0.$$

Differentiate the wave equation with respect to x to get

$$u_{xt} + au_{xx} = 0.$$

Combine the two results to get

$$u_{tt} = a^2 u_{xx}$$

which will replace the leading order error term, thereby providing with the modified equation.

(b) Consider the scheme

$$\frac{u_j^{n+1} - \left(u_{j+1}^n - u_{j-1}^n\right)}{\Delta t} + a\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0.$$

Using von Neumann approach, determine the stability condition.

(c) Using Taylor expansion, find the leading order error terms, and convert them to derive the modified differential equation of the above scheme.

(d) Find the dispersion relation of the modified equation.

From Lec#6 (Pg 9-10)

$$G(\lambda \Delta t) = 1 + \lambda \Delta t \quad \text{for Euler explicit.}$$

Where λ is the eigen value.

and $G = \dfrac{u_j^{n+1}}{u_j^n}$

Using $u(x,t) = e^{i(kx + \omega t)}$

i.e. $u(x_j, t_n) = u_j^n = e^{i(k x_j + \omega t_n)}$

find $\lambda = -\frac{a}{\Delta x}(1 - e^{-ik\Delta x})$

Also follow Heat eq^n example (Lec#6)

write $u_j^{n+1} = u_j^n - a\frac{\Delta t}{\Delta x}(u_j^n - u_{j-1}^n)$

This gives $G = 1 - a\frac{\Delta t}{\Delta x}(1 - e^{-ik\Delta x})$

We can now show that $\frac{a\Delta t}{\Delta x} \leq 1$

Ex: Consider RK4 method for 1D advection using backward in space scheme if $a > 0$.

Show that RK4 is stable for $CFL = \frac{a\Delta t}{\Delta x} \leq 1.39$

Ex: Repeat the above for centered in space and show that RK4 is stable for $CFL = \frac{a\Delta t}{\Delta x} \leq \sqrt{8} \approx 2.828$

Hint

$$\lambda = -ia\frac{\sin k\Delta x}{\Delta x}$$

$$G(\lambda\Delta t) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}, \quad z = \lambda\Delta t$$

$$= 1 - iy - \frac{y^2}{2} + \frac{iy^3}{6} + \frac{y^4}{24}$$

$$|G| = \left(1 - \frac{y^2}{2} + \frac{y^4}{24}\right)^2 + \left(y - \frac{y^3}{6}\right)^2$$

$$= \frac{y^6(y^2-8)}{576} = 0 \Rightarrow y = \sqrt{8}$$

$$y = \mathrm{Im}(\lambda\Delta t) = -\frac{a\Delta t}{\Delta x}\sin(k\Delta x)$$

$$|y| = \frac{a\Delta t}{\Delta x} \leq \sqrt{8}$$

$$\mathrm{CFL} = 2.828$$

Order of convergence of single-step methods

Consider $u(x_j, t) = \begin{bmatrix} u(x_1, t) \\ u(x_2, t) \\ \vdots \\ u(x_N, t) \end{bmatrix} = \vec{u}(t)$ or $U(t)$

as N-dimensional vector on N grid points $\{x_j\}_{j=1}^{N}$

Let $\vec{u}(t_n)$ be the exact solution at $t = t_n$

and $\vec{u}^{\,n}$ be the numerical solution at $t = t_n$

For some $p \in \mathbb{N}$ and $c > 0$, the error converges with order p: $\max_n \| \vec{u}^n - \vec{u}(t_n) \| \leq c \Delta t^p$

In other words

$$\max_{n,j} \| u_j^n - u(x_j, t_n) \| \leq c \Delta t^p$$

for fixed Δx.

This convergence is on time variable only,

p is known by the choice of scheme,

c is independent of choice of Δt, but

c is unknown.

Hence the convergence theorem does not tell much about the convergence rate as well as the ~~a~~ effect of Δx.

This is often known as method of line convergence.

Consistency of discretized PDE

Question that we may ask:

How much additional effort do we have to invest for a desired gain in accuracy.

Effort: the number of times we have to evaluate spatial operator accurately.

The cost of spatial discretization is given by the matrix-vector multiplication

$$\vec{u}(t) \longmapsto A\vec{u}(t)$$

where A denotes spatial discretization scheme

If there are N grid points, the cost is $O(N^2)$, which may be optimized to $O(N)$.

Hence the effort is like $\frac{1}{\Delta x}$ or $\frac{1}{\Delta x^2}$

Gain :

Assume

$$\varepsilon(\Delta t) = c\,\Delta t^{p}$$

The reduction of error by a factor $\rho > 1$

we want

$$\frac{\varepsilon(\Delta t_{old})}{\varepsilon(\Delta t_{new})} \approx \frac{c\,\Delta t_{old}^{p}}{c\,\Delta t_{new}^{p}} = \rho$$

$$\Rightarrow \quad \Delta t_{new} = \rho^{-1/p}\,\Delta t_{old}$$

To reduce error by a factor of $\rho > 1$ will increase the complexity by a factor of $\rho^{1/p}$

H.W. plot $\rho^{1/p}$ vs ρ to indicate second order method will have the best gain.

Examine the gain in accuracy for Euler explicit scheme using backward in space discretization

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c\,\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0$$

$$u_j^{n+1} = u_j(t_n + \Delta t) = u_j^n + \Delta t \left.\frac{\partial u}{\partial t}\right|_j + \frac{\Delta t^2}{2!}\left.\frac{\partial^2 u}{\partial t^2}\right|_j + \cdots$$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\partial u}{\partial t} + \frac{\Delta t}{2!}\frac{\partial^2 u}{\partial t^2} + \cdots$$

$$u_{j-1}^n = u^n(x_j - \Delta x) = u_j^n - \Delta x\,\frac{\partial u}{\partial x} + \frac{\Delta x^2}{2!}\frac{\partial^2 u}{\partial x^2} + \cdots$$

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{\Delta x}{2!}\frac{\partial^2 u}{\partial x^2} + \cdots$$

Hence

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c\,\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} + \frac{\Delta t}{2!}\frac{\partial^2 u}{\partial t^2} + \cdots - \frac{\Delta x}{2!}\frac{\partial^2 u}{\partial x^2} + \cdots$$